

StockMaster

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1. Introduction

1.1 Purpose: Define functional and non-functional requirements for StockMaster, a desktop-based inventory management system for small to medium-sized businesses.

1.2 Scope: Streamlined interface for managing products, suppliers, and categories using a persistent SQLite database for offline accessibility and data privacy.

2. Functional Requirements

2.1 Product Management

- FR-01: Create products (Name, Category, Price, Stock, Supplier).
- FR-02: Read/view products in tabular format.
- FR-03: Update product details.
- FR-04: Delete products.
- FR-05: Search by name or category.

2.2 Category & Supplier Management

- FR-06: Manage Categories (CRUD).
- FR-07: Manage Suppliers with contact details (Email, Phone).

2.3 Dashboard & Analytics

- FR-08: KPIs (Total Stock, Total Inventory Value, Low Stock Alerts < 5).
- FR-09: Bar Chart for product distribution.

2.4 Reporting

- FR-10: Generate PDF Inventory Report.
- FR-11: Include timestamp and summary in PDF.

3. Non-Functional Requirements

3.1 Usability (UI/UX)

- NFR-01: Modern Dark Theme.
- NFR-02: Intuitive Sidebar navigation.

3.2 Performance

- NFR-03: Search results in < 200ms for up to 10,000 items.
- NFR-04: Lightweight SQLite operations.

3.3 Reliability & Persistence

- NFR-05: Data persisted in SQLite.
- NFR-06: Database survives restarts/reboots.
- NFR-07: Auto-detect AppData for storage.

3.4 Compatibility

- NFR-08: Windows 10/11 support.
- NFR-09: Installer with desktop and Start Menu shortcuts.

4. Technical Stack

Language: Java 17+ (JDK 24 Compatible)

GUI Framework: JavaFX

Database: SQLite (via JDBC)

Build Tool: Maven

PDF Engine: Apache PDFBox

Installer: jpackage (WiX Toolset)

5. Future Enhancements (Roadmap)

- User Authentication (Login/Register)
- Cloud Sync / Multi-user support
- Barcode Scanner integration
- Sales tracking & Invoice generation